and hence

III.

Relative position is , with unit vector . Particles are confined to their tubes; their net forces are the electrostatic components along their motion. Their equations of motion are

Subtraction gives

Here is the acceleration of relative to . By , relative to moves as a particle of mass in a fixed-centre Coulomb field, with potential energy

Angular momentum and energy are conserved in a central field. Both refer to relative to ; the initial conditions give

The closest distance is ; there the velocity is perpendicular to , giving